A flexible modelling framework for estimating thermal tolerance and sensitivity

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Abstract

1. The thermal load sensitivity (TLS) framework is the gold standard for estimating and predicting thermal tolerance in nature. However, its two-step pipeline — first inferring median times to death across a range of assay temperatures, then regressing $\log_{10}$ time to death on temperature — rests on statistical assumptions that are routinely violated and that bias every downstream quantity (z, CTmax, heat-injury predictions).
2. We describe a flexible Bayesian, single-stage four-parameter logistic (4PL) model for modelling the impact of temperature and time on survival and sublethal responses to heat stress. We demonstrate how such an approach can recover all the classical TDT output typical of a two-stage modelling framework — including thermal sensitivity (z, slopes), critical thermal limit ($CT_{max_{1hr}}$) and the heat-injury (HI) integral, which allows for one to predict thermal injury in nature under realistic exposure temperatures. Given its fully Bayesian implementation, it allows for error propagation to account for uncertainty at all stages of modelling as well as improved inferences.
3. We applied our new modelling framework to simulated data, as well as case studies in brown shrimp (*Crangon crangon*), zebrafish (*Danio rerio*), and XXX to demonstrate the equivalencies and benefits of such an approach.
4. We show that the classical and Bayesian frameworks are equivalent, but that the hierarchical 4PL delivers properly propagated uncertainty, can account for complex hierarchical data (i.e., random effects), can accommodate censoring and overdispersion, and unlocks outputs the two-step pipeline cannot provide (full survival surface, arbitrary mortality percentiles, decomposable between-group contrasts). We provide reusable R helpers and recommend the hierarchical 4PL as the default analysis for new data using the TLS framework.

Introduction

Climate change is driving more frequent and extreme heat events globally, compromising organism fitness (Arnold *et al.* 2025; Rezende *et al.* 2020; Snook *et al.* 2026), with cascading consequences for population persistence and biodiversity (Smale *et al.* 2019; Wiens 2016). Quantifying differences in thermal tolerance across species is now central to forecasting climate change impacts on organisms in nature (Deutsch *et al.* 2008; Jørgensen *et al.* 2022; Pinsky *et al.* 2019; Sunday *et al.* 2011, 2012). Threshold-based measures of heat tolerance, such as CT_{max}, have long been used to establish differences in climate vulnerability among taxa and predicting climate change risk (Deutsch *et al.*

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2008; Pinsky *et al.* 2019; Pottier *et al.* 2025; Sunday *et al.* 2011, 2012). While useful, threshold-based measures do not capture how the negative effects of extreme heat on fitness depend on both the duration and intensity of heat stress, limiting our ability to make realistic predictions under fluctuating temperature regimes that occur in nature (Jørgensen *et al.* 2021; Ørsted *et al.* 2022; Rezende *et al.* 2014).

The Thermal Load Sensitivity (TLS) framework (or equivalently Thermal Death Time framework) (Arnold *et al.* 2025; Jørgensen *et al.* 2019, 2022; Noble *et al.* 2026; Ørsted *et al.* 2022; Rezende *et al.* 2014) resolves these limitations by jointly modelling the effects of heat intensity and duration on fitness (e.g., survival, fertility, reproduction). This matters because physiological damage from heat stress accumulates in a dose-dependent manner — physiological damage accumulates faster at higher temperatures and over longer exposures (Arnold *et al.* 2025; Jørgensen *et al.* 2019; Ørsted *et al.* 2022; Rezende *et al.* 2014). Organisms experiencing fluctuating thermal environments in nature can therefore accumulate damage to physiological systems that results in sublethal injury that compounds across successive heat exposures if not fully repaired (Arnold *et al.* 2025; Buckley *et al.* 2025; Jørgensen *et al.* 2021). Such processes can lead to delayed responses to extreme heat events.

Thermal load sensitivity modelling, while powerful, traditionally involves a two-step modelling approach (Jørgensen *et al.* 2019; Li *et al.* 2023; Ørsted *et al.* 2022; Rezende *et al.* 2020; Truebano *et al.* 2018). First, experiments will typically be expose organisms to a range of temperatures for various durations of exposure. The proportion of live/dead, fertile/infertile, or live/knockdown, for instance, is then modelled as a function of log transformed exposure time (in minutes or hours) using dose-response curves fit across all temperatures (usually at each temperature treatment independently). The time to death, fertility, or heat knockdown is then derived from these curves, based on the temperature predicted to reduce these responses by a given threshold (e.g., t_{50}, the time to reach 50% mortality). Second, the log10 transformed t_{50} and temperature are regressed against each other using a linear model. Thermal sensitivity (i.e., z, the inverse of the slope of this regression) and static critical thermal maxima (i.e., the intercept of the regression, $sCT_{max_{time}}$) are then derived from this linear model (Ørsted *et al.* 2022; Rezende *et al.* 2014). $sCT_{max_{time}}$ is usually computed for a 1 min (Rezende *et al.* 2014), 1 hour (Jørgensen *et al.* 2019), or 4 hour (Ørsted *et al.* 2024) duration of exposure to heat stress. The 1 hour reference falls within typical assay durations avoiding substantial extrapolation which is why it is often recommended (Jørgensen *et al.* 2019). Thermal sensitivity and $CT_{max_{1hour}}$ are than used to quantify Heat Injury (HI) (e.g., Jørgensen *et al.* (2019); Villeneuve & White (2024)) in response to realistic temperatures in nature. Heat injury measures the cumulative percentage or probability until reaching a given threshold (e.g., t_{50}), at which point we would predict 50% of the population to have died. Recent work has also emphasized the importance of repair in heat injury dynamics (Arnold *et al.* 2025; Buckley *et al.* 2025), and new modelling approaches have recently been developed to capture these damage-repair dynamics (Arnold *et al.* 2025).

While the two-step approach to modelling thermal sensitivity is intuitive and simple in practice, it has numerous practical and statistical limitations. First, the two-step pipeline does not typically allow correct uncertainty propagation. The t_{50} point estimates have non-negligible sampling variance. Treating them as fixed inputs to the regression discards that sampling variance, which can lead to miscalibrated standard errors on the slope (z) and intercept (CT_{max}), and inflated R^2 – a known issue in the statistical literature (Burke *et al.* 2017; Carroll *et al.* 2006). Second, group comparisons (between species, sexes, or life stages) inherit this miscalibrated uncertainty. As such, differences in z between groups cannot be attributed to differences in the upper asymptote, lower asymptote, or curve steepness of the dose-response curve, because all three are collapsed into a single t_{50} without sampling variance before the comparison is made. Third, the t_{50}-extraction step collapses the within-replicate count, and prevents the hierarchical decomposition of variance across replicates, plates, populations, sexes, and life stages that could affect parameters of dose-response curves (Harrison 2014; Harrison *et al.* 2018; Nakagawa & Santos 2012). Fourth, the intrinsically hierarchical structure of the experimental data(e.g., animals within batches and days of experiments) is largely ignored, which can lead to pseudoreplicaton, resulting in anticonservative standard errors that affect biological inference. Finally, propagating uncertainty from a two-step fit into downstream predictions — such as cumulative heat injury — is rarely attempted in practice understating the credible range of extreme heat impacts in nature.

Here, we present an alternative to the two-step pipeline: a single hierarchical four-parameter logistic (4PL) model fit jointly across all temperature by exposure duration data. The 4PL is a flexible sigmoidal form that pairs with different sampling distributions — binomial for counts of dead or infertile individuals, time-to-event for latencies (e.g., time to knockdown), Beta for continuous proportions, or Gaussian for continuous performance measures (Nielsen *et al.* 2004; Ritz 2010; Ritz *et al.* 2015; Ritz & Streibig 2008; Rudemo *et al.* 1989). We focus on binomial outcomes for clarity, since these dominate the TLS literature (Ørsted *et al.* 2022, 2024; Rezende *et al.* 2020), but the derivations below apply unchanged when the likelihood is swapped. Additionally, while we use the four-parameter log-logistic form throughout for its flexibility other monotonic sigmoidal families (e.g., Weibull, log-normal) and simpler nested versions of the log-logistic (3PL, with one asymptote fixed; 2PL, with both fixed at 0 and 1) can be used within this framework (Ritz *et al.* 2015; Ritz & Streibig 2008). We show how all the classical thermal load sensitivity parameters (z, CT_{max}) can be derived from this single model, while also easily allowing for predictions of heat injury accumulation, survival, and their uncertainty in fluctuating thermal regimes. We develop these models within a Bayesian fr amework which allows for better uncertainty calibration and error propagation with complex hierarchical data structures.

A non-linear Bayesian hierarchical model for characterizing thermal load sensitivity

Model specification

Assuming we have mortality data for different times and temperatures, let y_{ij} denote the number of survivors out of n_{ij} individuals in trial j exposed for time t_{ij} (e.g., minutes) at assay temperature T_{ij} (°C). We can model the counts directly with a binomial likelihood (but a beta-binomial likelihood function can also be used and controls for overdispersion):

$$y_{ij} \mid n_{ij}, p_{ij} \;\sim\; \text{Binomial}(n_{ij},\, p_{ij}), \tag{1}$$

with survival probability p_{ij} given by a four-parameter logistic (4PL) function of $\log_{10}$-exposure time,

$$p_{ij} \;=\; \text{low} \;+\; \frac{\text{up} - \text{low}}{1 + \exp\!\big(k\,(\log_{10} t_{ij} - mid_{ij})\big)}. \tag{2}$$

where low is the lower survival probability asymptote, up is the upper survival probability asymptote, and k is the slope defining the change in survival probability on the $\log_{10} t$ axis. Larger k gives a sharper decline in survival probability with $\log_{10} t$. mid_{ij} is the value of $\log_{10} t$ at which p falls to the midpoint between low and up. When asymptotes are near 0 and 1, mid_{ij} coincides with $\log_{10} t_{50}$ at trial j. While other logistic forms can be used (e.g., two or 3 parameter logistics), Equation 2 is the most flexible model that can account for different asymptotic survival probabilities.

The key parameters of Equation 2 (i.e., low, up, k, mid_{ij}), can be fit as a non-linear function within a likelihood or Bayesian framework, which estimates their correlation and allows for fixed or random effects to be estimated to explain variation in these parameters. Such an approach allows for *a priori* comparisons (e.g., do sexes or species differ in midpoints?), with correct inference (e.g., do species differ significantly from each other?) and allows for variance decomposition to help identify key sources of variance in the data (e.g., estimating variances explained across individuals, species, trials etc.). Such models are most flexibly fit within a Bayesian framework with intuitive interpretation of these parameters and a highly flexible path for error propagation. We show below how such a model can be used to derive all thermal load sensitivity parameters of interest to ecophysiologists.

Deriving key thermal load sensitivity parameters

The TLS framework is elegant in that it derives metrics with clear biological interpretation that can be compared across populations and species (e.g., z, $CT_{max_{1hr}}$). Here, we show that we can derived these same metrics directly from Equation 2.

As indicated above, parameters can be fit to model changes in the mean of any of the four parameters in Equation 2. We can easily derive thermal sensitivity (z) and $CT_{max_{1hr}}$ by modelling the effect of temperature on midpoints as follows:

$$\text{mid}_{ij} = \beta_0 + \beta_1\,(T_{ij} - \bar{T}), \tag{3}$$

where mid_{ij} is the midpoint ($\log_{10}$-time at 50% survival between low and up) for trial j at assay temperature T_{ij} (°C); β_0 is the intercept giving the midpoint at the grand mean temperature $\bar{T}$; β_1 is the slope describing how the midpoint changes per °C increase in T_{ij}; and $\bar{T}$ is the grand mean of assay temperatures used to mean-centre T_{ij}, which decorrelates β_0 from β_1 and improves parameter estimation and interpretation (Schielzeth 2010).

Setting $p_{ij} = 0.5$ in Equation 2, isolating the exponential, and taking natural logs gives $\log_{10} t_{50}(T) = \text{mid}(T) + (1/k)\log[(\text{up} - 0.5)/(0.5 - \text{low})]$. Substituting into Equation 3 and grouping all terms that do not depend on T into an intercept, α, recovers the classical TDT linear relation:

$$\log_{10} t_{50}(T) \;=\; \underbrace{\left[\beta_0 + \tfrac{1}{k}\log\tfrac{\text{up}-0.5}{0.5-\text{low}} - \beta_1\bar{T}\right]}_{\alpha} + \beta_1\,T, \tag{4}$$

Note that Equation 4 subtracts $\beta_1\bar{T}$ in α because we mean-centred T. With this subtraction, the intercept estimates $\log_{10} t_{50}$ at the average temperature ($\bar{T}$); without it, the intercept estimates $\log_{10} t_{50}$ at $T = 0$°C. From Equation 4, the classical TLS quantities fall out directly. Thermal sensitivity is $z = -1/\beta_1$, and the critical thermal limit at any reference time t_{ref} is $CT_{max}(t_{\text{ref}}) = (\log_{10} t_{\text{ref}} - \alpha)/\beta_1$ (with $t_{\text{ref}} = 60$ min for $CT_{max_{1hr}}$). Each of these quantities can be estimated separately for different strata (e.g., sex, population, species) by including the relevant fixed covariates and/or random effects in the linear predictor for mid_{ij} — and, if appropriate, in the linear predictors for low, up, and k, each of which is modelled separately in the joint Bayesian fit (see Equation 12 below).

The same derivation generalises beyond the median: the Bayesian dose-response fit gives access to *any* mortality percentile, not just t_{50} (Jørgensen *et al.* 2021; Rezende *et al.* 2020). Setting $p_{ij} = x/100$ in Equation 2 and following the same algebra used for t_{50}, the posterior samples of $(\text{low}, \text{up}, k, m(T))$ deliver the time at which $x\%$ of the population reaches the assayed endpoint:

$$\log_{10} t_x(T) \;=\; m(T) \;-\; \frac{1}{k}\,\ln\frac{\text{up} - x/100}{x/100 - \text{low}}, \tag{5}$$

Varying x in Equation 5 then yields any threshold of interest. For instance, t_{10} marks the onset of early mortality — useful for risk management — while t_{90} is closer to a population-collapse threshold. As with t_{50}, every prediction inherits the joint posterior, so uncertainty bands can be reported alongside the population-level median.

Importantly, the slope β_1 — and hence z — is robust to estimated asymptotes because the asymmetry correction $\frac{1}{k}\log\frac{\text{up}-0.5}{0.5-\text{low}}$ does not depend on T. Instead, it shifts $\log_{10} t_{50}$ vertically without changing the slope. In contrast, threshold summaries are not robust in this way; when partial mortality or sparse data do not estimate the asymptotes to be around 0 and 1, the asymmetry correction must be retained — added to the naive $\log_{10} t_{50}$ (Equation 6) and subtracted, after division by the slope (to convert the vertical shift in $\log_{10} t_{50}$ into the corresponding shift in temperature), from the naive CT_{max} (Equation 7):

$$\log_{10} t_{50}(T) \;=\; \underbrace{\beta_0 + \beta_1(T - \bar{T})}_{\text{naive estimate}} \;+\; \underbrace{\tfrac{1}{k}\log\tfrac{\text{up}-0.5}{0.5-\text{low}}}_{\text{correction}}, \tag{6}$$

$$CT_{max}(t_{\text{ref}}) \;=\; \underbrace{\frac{\log_{10} t_{\text{ref}} - \beta_0 + \beta_1\bar{T}}{\beta_1}}_{\text{naive estimate}} \;-\; \underbrace{\tfrac{1}{\beta_1}\cdot\tfrac{1}{k}\log\tfrac{\text{up}-0.5}{0.5-\text{low}}}_{\text{correction}}. \tag{7}$$

In practice, however, these corrections may not be needed so long as the experimental design includes treatments that span the full mortality range which anchors low and up tightly near 1 and 0 and makes the correction term effectively zero. When this is the case, $\log_{10} t_{50}$ and CT_{max} can be read directly from the naive forms in Equation 6 and Equation 7. Regardless, we recommend using the correction factor as a safeguard. However, computing both the naive and corrected estimates and comparing them is a useful sensitivity check for whether the correction is needed in any given dataset.

Predicting heat injury and survival under dynamic thermal environments

A particularly powerful aspect of TLS models are their ability to translate dynamic temperature times series' from nature to predict heat injury (HI) accumulation and thus population survival probability (Jørgensen *et al.* 2021; Li *et al.* 2023; Ørsted *et al.* 2022, 2024; Rezende *et al.* 2020). Heat injury accumulates as the time spent at a given temperature divided by the median lethal time at that temperature, integrated across the temperature trajectory $T(i)$ as follows:

$$HI(t) \;=\; 100\int_0^t 10^{\left(T(i)-CT_{max_{1\mathrm{hr}}}\right)/z}\,di, \tag{8}$$

where z is the thermal sensitivity and $CT_{max_{1\mathrm{hr}}}$ is the static temperature at which 50% of the population reaches the assayed endpoint after one hour of exposure. When $HI(t) = 100\%$, the cumulative dose equals one t_{50} at the 1-hour reference, predicting 50% mortality (or 50% loss of the assayed sublethal endpoint, e.g. fertility; Ørsted *et al.* (2024)). Injury accumulates only when $T(i)$ exceeds a critical temperature T_c below which damage is repaired as fast as it accrues (Jørgensen *et al.* 2021; Ørsted *et al.* 2022). Field temperature records are typically available at hourly resolution. Equation 8 can then be evaluated by summing $100 \cdot 10^{(T_i - CT_{max_{1\mathrm{hr}}})/z}$ across each hourly interval whose temperature T_i exceeds T_c, following the implementations in Jørgensen *et al.* (2021), Ørsted *et al.* (2024), and Li *et al.* (2023). To propagate uncertainty, z and $CT_{max_{1\mathrm{hr}}}$, calculated from our Bayesian dose-response model can be used to calculate $HI(t)$ through time by substituting samples from the posterior distribution for each parameter into Equation 8.

A more natural prediction to make may instead be the population-level survival probability over time. This has the advantage of allowing for a direct connection to population dynamical models which typically use survival, growth and fecundity as vital rates for population-level predictions (Noble *et al.* 2026). To predict survival, we can calculate the *effective exposure time* at a chosen reference temperature T_{ref}. More effective time is accumulated at hot temperatures and less at cool ones, in proportion to the scaling factor $10^{(T(i)-T_{\mathrm{ref}})/z}$,

$$\tau_{\mathrm{eff}}(t) \;=\; \int_0^t 10^{\,(T(i)-T_{\mathrm{ref}})/z}\,di, \tag{9}$$

We can then evaluate the 4PL surface once at the cumulative effective exposure to obtain the predicted population-level survival fraction at every point in the trajectory:

$$S_{\mathrm{pred}}(t) \;=\; S\big(T_{\mathrm{ref}},\, \tau_{\mathrm{eff}}(t)\big). \tag{10}$$

Here $S(T,\tau)$ is the survival surface fitted by the dose-response model, so $S_{\mathrm{pred}}(t)$ is the survival probability for an organism held at *constant* T_{ref} after exposure duration $\tau_{\mathrm{eff}}(t)$.

Application of Bayesian dose-response models to the TLS framework

We used a four-parameter logistic function to model the impact of temperature and duration (hours/minutes) on survival probability thermal sensitivity, estimate CT_{max1h}

To break the well-known likelihood ridges among (low, u, k) in unconstrained 4PL fits (Bolker 2008; Gelman 2006), we estimate the identifiable parameterisation

$$\mathrm{low} = 0.49\,\mathrm{logit}^{-1}(\widetilde{\mathrm{low}}), \qquad \mathrm{up} = 0.51 + 0.49\,\mathrm{logit}^{-1}(\widetilde{\mathrm{up}}), \qquad k = \exp(\kappa), \tag{11}$$

with $\widetilde{\mathrm{low}}, \widetilde{\mathrm{up}}, \kappa \in \mathbb{R}$. This constrains $\mathrm{low} \in [0, 0.49]$ and $\mathrm{up} \in [0.51, 1]$ into disjoint intervals so that $u > \mathrm{low}$ by construction, the curve direction is unambiguous, and k is positive on the natural scale.

The inflection point m_{ij} is modelled as a linear function of (mean-centred) assay temperature, with additional fixed covariates and nested random effects:

$$m_{ij} \;=\; \beta_0 \;+\; \beta_1 \, (T_{ij} - \bar{T}) \;+\; \mathbf{x}_{ij}^{\top}\gamma \;+\; b_j \;+\; \varepsilon_{ij}, \tag{12}$$

where $\bar{T}$ is the grand mean of assay temperatures (centring stabilises sampling and decorrelates β_0 from β_1), $\mathbf{x}_{ij}$ holds any additional fixed covariates with coefficients γ, b_j collects the nested random effects and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is an observation-level random effect that absorbs the binomial overdispersion characteristic of replicated count assays (Harrison 2014). However, we could also remove the observation-level random effect from the model and instead fit a beta-binomial error distribution which estimates a dispersion parameter (ϕ) more naturally.

Together, Equation 1–Equation 12 define the hierarchical Bayesian 4PL fit directly to the raw binomial counts. Every classical TDT quantity — slope, z, CTmax at any reference time, and the heat-injury integral — is recovered as a one-line posterior summary of this single joint fit, as derived in §[Equivalence with the classical pipeline].

Case Study 1: Thermal sensitivity estimation across life stages in shrimp

What the 4PL delivers that the two-step cannot

Worked example: brown shrimp thermal tolerance

Data and experimental design

Lethal TDT — hierarchical 4PL on counts

Sublethal time-to-death

Dynamic CTmax across ramping rates

Reconciling the three endpoints via the Jørgensen ramping equation

Field-temperature heat-injury projections under warming scenarios

Discussion

What changes for the field if the hierarchical 4PL becomes the default

Limits of the present analysis

Recommendations for primary studies and meta-analyses

Conclusions

Acknowledgements

Author contributions

Data and code availability

All data, analysis scripts, and the rendered manuscript and supplement are available at https://github.com/daniel1noble/tls_model_equivalence.

Conflict of interest

The authors declare no competing interests.

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